

SOFTWARE DEVELOPMENT PROCESSES

Lab Sheet 3 – Design

INTRODUCTION

This week in lectures, we talked about software design. Software design is a large and varied discipline, and there is often no one right way to design a system. Those of you studying the Systems Analysis & Design unit this term will be looking at some common design patterns, used to overcome common problems seen in system design. Our interest in Software Development Processes, however, lies firmly in the *architecture* of the system. Understanding what the main components of a system should be allows for working groups to be formed, allows for a schedule to be planned etc.

TASK 1: IDENTIFYING SYSTEM COMPONENTS

Most meaningfully-sized systems are made up of a number of smaller components. With experience, you will begin to develop a sense of how to break a system down into components based on an initial description. When learning, however, a good starting point is to ask yourself “Which bits of functionality need to take place where?” and “Where will the data be kept?”. The answers to these two questions often constrain you to a fairly small number of possibilities. Discuss your answers to these questions with the people around you.

Sketch an informal block diagram depicting your proposed design for the system. Use a simple labelled rectangle on the diagram for each component, and use arrows between the rectangle to illustrate which components will need to communicate with which others.

TASK 2: COMMUNICATION & COORDINATION

Looking at your block diagram from Task 1, take a look at the arrows between components. What operations do these arrows represent? Is it one class in a program calling a method from another class? Is it a file transfer? Is it a web service client making a request of a server? Give some examples of the methods & parameters/data formats that would be involved. Show your examples to the people around you, and see whether you can explain to them how the components would work together.

Try to sketch out a short storyboard of a particular operation happening on the system, starting with a user and following the chain of events through the various components in your system design. Note which component the user initially interacts with, and trace the communication between the various components for this operation.

TASK 3: DATA DESIGN

It is likely that at least one of the components in your system design is some kind of data store or database. Sketch out a copy of a schema for the data store in an appropriate format. What data will you need to keep? What would the appropriate data types be? What would the relationships between the entities be?

TASK 4: CLASS DISCUSSION

Your lab tutor will gather everyone together with time before the end of the lab session to go through your designs. Did you use a similar architecture to the other people in your class? Do people have a similar vision for the system's operation? Would other people's designs be viable? Did people have similar thoughts on what data will need to be stored, how and where?

Keep a copy of your finalised system architecture block diagram, this will be helpful in future weeks.